W-C-421H
November 2, 1987
TO SUPERSEDE
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May 17, 1973

FEDERAL SPECIFICATION

CLEANER, VACUUM, ELECTRIC (HOUSEHOLD/COMMERCIAL)

This specification has been approved by the Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1. Scope. The cleaners covered by this specification are canister, upright, and combination models for household and commercial duty. These cleaners are designed to operate on 120 VAC, 60 Hertz or 220 VAC, 50 Hertz, single phase power source as specified (see 6.2) and are to be used for dry pickup only.

1.2 Classification.

1.2.1. Types, classes, and sizes. The cleaners to be furnished in accordance with this specification shall be of the following types, classes, and sizes (as applicable):

Type I - Canister.
  Class A - Household.
  Class B - Commercial.

Type II - Upright.
  Class A - Household.
  Class B - Small commercial.
  Class C - Large commercial.
    Size 1 - 12 inch (minimum) wide agitator.
    Size 2 - 16 inch (minimum) wide agitator.

Type III - Combination.
  Class A - Household.

2. APPLICABLE DOCUMENTS

2.1. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues used shall be those in effect on the date of the solicitation or request for bids.

FSC 7910

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2.2 In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards are available from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120)

Underwriters' Laboratories (UL), Inc., Standard:

UL Standard 1017 - Vacuum Cleaning Machines and Blower Cleaners.

(Copies of UL Standards are available from Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, Illinois 60062)

American Society for Testing and Materials (ASTM) Standards:

ASTM Standard F 558 - Method for Measuring Air Performance Characteristics for Vacuum Cleaners.

(Copies of ASTM Standards are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103)

3. REQUIREMENTS

3.1 Material. The vacuum cleaners shall be composed of materials normally used in commercial products and shall be free from defects affecting the appearance, serviceability, or durability.

3.2 UL conformance. The vacuum cleaners shall conform to UL Standard 1017.

3.3 General. Each cleaner shall be complete, fully lubricated, and ready to operate; minor assembly is permitted. The design shall permit easy accessibility to component parts for inspection, adjustment, cleaning, or replacement, and shall provide reliable operation. All cleaners of one type, class, size, and model furnished under this CID for any one contract, including parts and assemblies thereof, shall be interchangeable. Interchangeability shall not require alteration or modification.

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3.3.1 Type I (Canister). Each type I cleaner shall consist of a housing, handle or grip, vacuum producer assembly, filter assembly, switch, cord assembly, tools and attachments as specified (see 3.4.5).

3.3.2 Type II (Upright). Each type II cleaner shall consist of a housing and handle assembly, vacuum producer assembly, nozzle and agitator assembly, filter assembly, switch, and cord assembly. Tools and attachments shall be furnished when specified (see 3.5.6 and 6.2).

3.3.3 Type III (Combination). Each type III cleaner shall consist of the type I components plus a powered agitator assembly to be affixed to the end of the hose and wand assembly. The cleaner shall conform to the requirements for type I, class A, the agitator shall conform to the applicable requirements of type II, class A.

3.3.4 Finish. All exterior surfaces, all interior surfaces impinged upon by the airflow, and all surfaces subject to contact by the operator, shall be smooth and free from surface defects or obstructions which may contribute to dirt adherence or operator injury. Exterior metallic surfaces shall be smooth, of uniform appearance, and suitably plated or finished if necessary to resist abrasion and corrosion. Platings and other finishes shall be in accordance with the best commercial practice.

3.3.5 Portability. Handles or grips capable of supporting the weight of the machine shall be furnished.

3.3.6 Stability. The cleaners shall conform to the stability requirements of UL 1017.

3.3.7 Housings. The housings on all cleaners shall be strong, lightweight, and of good appearance. The housing shall be encircled on the front and sides by one or more resilient, non-marking guards or bumpers to protect surfaces and furniture.

3.3.8 Vacuum producer. The motor and blower assembly shall be of rugged construction and balanced so there is minimum vibration when in operation. If the airflow generated by the blower is used to cool the motor, the air shall be adequately filtered before entering the motor.

3.3.9 Filter assembly. The filter assembly of each cleaner shall be of adequate capacity to allow passage of air through the filter at the full flow of the vacuum producer and shall employ disposable or easily cleaned dirt receptacles. All joints in the collection/filtration system shall be dust tight. All filters and dust receptacles, including motor intake filters, shall be adequately fitted and secured to prevent loosening during use, and shall be easily accessible for cleaning or replacement.

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3.3.10 Switch(s). The switch(s) shall be in accordance with UL Standard 1017.

3.3.11 Power cord assembly. Class A cleaners shall have a power cord of 20 foot minimum; Class B and C cleaners shall have power cords of 30 foot minimum. All Class B and C cleaners, except for double insulated cleaners, shall be provided with a 3 conductor power cord assembly conforming to the applicable requirements of UL Standard 1017. All power cords shall be provided with reinforcement or a strain relief at its point of entry into the housing or handle. The cord shall be measured from the point of entry into the housing or handle to the flat of the connector plug.

3.3.12 Cord storage. Each cleaner shall provide some means for coiling, wrapping, or otherwise storing the power cord when the cleaner is not in use.

3.3.13 Lubrication. The motor bearings employed in the cleaner shall be sealed or shielded and shall not require periodic lubrication. All lubricated points shall be adequately shielded to prevent dirt entrance and shall not permit excessive leakage of lubricants.

3.3.14 Tools and attachments. All tools and attachments provided with the cleaners shall be of non-ferrous alloy, suitably anodized or otherwise finished, or impact resistant non-metallic material. All corners shall be rounded and finished to protect furniture, fabrics, and operator.

3.3.15 Noise. The maximum noise generated by these cleaners while in operation shall be 85 dbA. Measurements shall be taken in accordance with 4.3.2.2.3.

3.3.16 Identification marking. Each cleaner shall be legibly marked with the manufacturers model number or part number and his name or trademark of such known character that the manufacturer shall be easily determined.

3.4 Type I (Canister).

3.4.1 General. The applicable requirements of 3.3 shall apply.

3.4.2 Housing. The housing of type I cleaners shall contain all components of the cleaner except for the accessories. The housing shall be equipped with wheels or casters. A dolly with provisions for attachment to the cleaner may be acceptable if the housing is not wheeled. The housing shall be designed such that the cleaner will operate satisfactorily when placed on a stair tread. Hand portability shall be provided by a handle or handles that are an integral part of, or securely fastened to, the housing.

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3.4.3 Performance. The type I cleaners shall be tested in accordance with 4.3.2.2. In the performance tests (4.3.2.2.4) the class A cleaners shall develop a minimum of 25 inches of water lift through a 3/4 inch orifice and 5 inches of water lift through a 1-1/4 inch orifice. The class B cleaners shall develop 35 inches of water lift through a 3/4 inch orifice and 9 inches of water lift through a 1-1/4 inch orifice.

3.4.3.1 Motor. The motors employed by the class A cleaners shall be rated for 1 hour, varying duty. The motors employed by the class B cleaners shall be rated for continuous duty. Continuous duty is defined as constant operation for an 8 hour work day. The nameplate data shall be in accordance with UL Standard 1017.

3.4.3.2 Blower. The blower shall be balanced, shall generate a minimum of noise, and shall be capable of developing the minimum water lift specified when tested in accordance with 4.3.2.2.4.

3.4.4 Filter assembly. The minimum capacity for the type I cleaners shall be 200 cubic inches. The class A cleaners shall utilize disposable dust bags. The class B cleaners may use disposable dust bags. The dust bag shall be secured to the cleaner outlet in such a manner that is dust tight and can be expected to hold the bag in place during normal use with the bag empty and with the bag filled. There shall be no sharp projections or edges in the area of dust bag expansion. Cleaners with disposable dust bags shall be supplied with a minimum of two (2) dust bags.

3.4.5 Tools and attachments. Each type I cleaner shall be provided with the following tools and attachments.

3.4.5.1 Hose. The hose assemblies for the type I cleaners shall be 6 feet minimum in relaxed length and 1-1/4 inch minimum in inside diameter. The hose may be plastic or any durable composition and shall be steel wire reinforced. The class B hose connection shall be positive locking. When tested as specified in 4.3.2.1.1, the hose shall not exhibit any rupture or permanent deformation greater than 1/4 inch. The couplings on the class B hoses shall be tested as specified in 4.3.2.1.2.

3.4.5.2 Wand. Each type I cleaner shall be provided with a one piece, two piece, or telescoping wand assembly of 1-1/8 inch minimum inside diameter. Overall assembled length shall be 42 inches minimum. The wands shall be fabricated in accordance with 3.3.14. The wands for the class B cleaners shall be positive locking. The bore for all wands shall be clean and smooth. The couplings on the class B wands shall be tested as specified in 4.3.2.1.2.

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3.4.5.3 Tools. Each cleaner shall be provided with a bare floor tool and a carpet tool (or combination), a crevice tool, an upholstery nozzle, and a brush. The tools shall be non-marking and fabricated from non-ferrous alloys or impact resistant synthetics. The brushes shall be natural or synthetic fibers densely packed in a holder or backing. The bare floor tool and the carpet tool (or combination bare floor/carpet tool) shall be designed to swivel 85 degrees minimum in both directions from the normal position with the tool properly affixed to the wand.

3.5 Type II (Upright).

3.5.1 General. The applicable requirements of 3.3 shall apply.

3.5.2 Housing. The housing of type II cleaners shall be of durable construction and shall contain the running gear to provide the mobility, the vacuum producer assembly, and an agitator. The handle shall be attached to the rear portion of the housing. The frame shall be of rugged construction and shall be enclosed in a shroud of metal or impact resistant synthetic.

3.5.2.1 Wheels. Wheels on the class B and class C cleaners shall be either bearing supported and permanently lubricated or self lubricating polymeric wheels. All wheels shall be non-marking.

3.5.2.2 Handle. The handle assembly of the class A cleaner shall be of metal, fiberglass, or similar durable synthetic. If the handle is metal the grip shall be composed of a synthetic material. The handle assembly of the class B and C cleaners shall be of metal tubing and shall have a synthetic grip. The handles of all type II cleaners shall be of adequate strength and rigidity to permit elevating and rotating the agitator assembly about the rear wheels without detectable flexure. The handle shall pivot smoothly from the vertical to a near-horizontal position. A detent or "park" position shall be provided to hold the handle in the vertical position for storage. Other detents or "work" positions are permitted. Provisions for cord storage shall be provided on the handle. The power switch may be located on the handle.

3.5.3 Performance. The cleaners shall be subjected to the tests of 4.3.2.2. In the performance tests (4.3.2.2.4) the cleaners shall develop a minimum of 10 inches of water lift through a 3/4 inch orifice and a minimum of 6 inches of water lift through a 1 inch orifice when tested.

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3.5.3.1 Motor. The motors employed by the class A cleaners shall be rated for 1 hour, varying duty. The motors employed by the class B cleaners shall be rated for continuous duty. Continuous duty is defined as constant operation for an 8 hour work day. The nameplate data shall be in accordance with UL Standard 1017. The agitators (see 3.5.5) may be run by a separate motor or may be driven by the vacuum producer.

3.5.4 Filter assembly. The minimum capacity for the class A cleaner dust bag shall be 200 cubic inches. The class A cleaners shall utilize a disposable dust bag. The minimum capacity for class B and C cleaner dust bag is 500 cubic inches. The class B and C cleaners may utilize disposable dust bags. All connections between the nozzle and the bag shall be dust tight. The bag may be enclosed in a shroud of plastic or suitable synthetic. Cleaners using disposable dust bags shall be furnished with a minimum of two (2) dust bags.

3.5.5 Nozzle and agitator assembly. The nozzle shall be of rugged metal or polymeric construction and shall effectively funnel litter from the surface being cleaned to the suction channel. The nozzle shall automatically adjust or shall be adjustable in height to compensate for varying pile depths. The agitator shall be provided with high quality bristles and may employ beaters. The bristles or the entire agitator shall be easily replaceable on class B and C cleaners. When tools and attachments are employed with the type II cleaners, a means shall be provided to automatically or manually uncouple the agitator. Manual uncoupling shall not require disassembly.

3.5.6 Tools and attachments. When specified (see 6.2), the type II cleaners shall be furnished with a hose, wand assembly, bare floor tool, upholstery nozzle, brush, and crevice tool. The type II, class A attachments and tools shall meet the requirements of the type I, class A cleaner (see 3.4.5). The type II, class B and C attachments and tools shall meet the requirements of the type I, class B cleaner (see 3.4.5).

3.6 Type III (Combination).

3.6.1 General. The applicable requirements of 3.3 shall apply. The requirements of type I, class A cleaners and type II, class A cleaners (as applicable) shall also apply.

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3.6.2 Powered agitator. The agitator shall be powered by an independent motor contained in the nozzle housing. The motor shall be rated a minimum of 1/10 HP and shall operate from the same power source provided by the power connection for the cleaner. The power cord for the agitator shall be securely clamped to or be molded into the suction hose/wand assembly. The power switch for the agitator may be located on the grip.

3.6.3 Tools and attachments. The type III cleaners shall furnished with tools and attachments as specified in 3.4.5.

3.7 Manuals. Each cleaner shall be furnished with a manual of operating instructions. The class B and C cleaners shall, in addition to the operating instructions, be furnished with a handbook of maintenance instructions with an Illustrated Parts Breakdown (IPB). The IPB shall identify all components by part number to the lowest repair level.

3.8 Workmanship. The cleaners and all tools and attachments shall be constructed, assembled, and finished to assure good quality equipment of a neat general appearance and free from imperfections and damage that may affect appearance, serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified in this specification. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections or tests set forth in the specifications where such action is deemed necessary to assure that supplies and services conform to prescribed requirements. The contractor is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and section 5 of this specification. The inspections set forth in this document shall become a part of the contractor's overall inspection system or quality control program. The absence of any inspection requirements in this or any referenced document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Component testing. In lieu of using his own facilities, the contractor may submit a report of analysis and performance of component materials from a laboratory acceptable to the Government, whether an independent testing facility or the laboratory of the component material supplier, as evidence of compliance with the specification requirements. The report shall include, in addition to the actual data, the manufacturer's name, the name and address of the testing laboratory, and the date of test.

4.3 Sampling for examination and tests. Samples for examination and tests shall be selected in accordance with MIL-STD-105, inspection level S-2, Acceptable Quality Level (AQL) 4.0 defects per hundred units. A lot shall be all vacuum cleaners of one type, class, and size (as applicable) submitted for inspection at one time. A sample unit shall consist of a vacuum cleaner complete with all tools and attachments (when specified).

4.3.1 End item inspection.

4.3.1.1 Visual examination. The sample cleaners shall be visually examined for the following specific defects. The cleaners and tools shall also be subjectively examined for workmanship (see 3.8).

1. No UL listing or other evidence of conformance to UL Standard 1017 requirements.
2. Not finished as specified (see 3.3.4).
3. Handles and grips not as specified (see 3.3.5).
4. Cleaner not stable (see 3.3.6).
5. Power cord not as specified (see 3.3.11).
6. Tools and attachments not as specified (see 3.3.14 and applicable requirements).
7. Manual is inadequate (see 3.7).
8. Dimensions not as specified.

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4.3.2. Tests.

4.3.2.1 Pre-operational tests.

4.3.2.1.1 Hose test. A compressive load of 150 pounds (minimum) shall be applied to any portion of the hose, at least 12 inches from any fitting, by means of parallel surfaces having a length of not more than 3 inches parallel to the axis of the hose for not less than 2 minutes. Upon removal of the weight there shall be no rupture and the hose diameter shall not differ more than 1/4 inch from the original diameter.

4.3.2.1.2. Coupling test, tools and attachments for class B and C cleaners. A tensile force of not less than 25 pounds shall be applied to the engaged hose, wand, and all tools for not less than 2 minutes each. The couplings shall not part during the test and shall engaged and release satisfactorily after the test.

4.3.2.2 Operational and UL conformance tests. Unless otherwise specified a certificate of conformance to these test requirements may be acceptable in lieu of testing each lot. The contractor shall have available for review evidence that those tests specified by this document and by the industry standards referenced herein have been performed, that the product tested met the requirements specified and that the product being offered has been manufactured using the same materials, processes, and manufacturing techniques, and is otherwise identical to the product tested.

4.3.2.2.1 UL tests. The cleaners shall be tested as specified by UL Standard 1017.

4.3.2.2.2 General operations. This operational test shall be performed with the cleaner connected to a power source conforming to the motor nameplate data. Each class A cleaner shall be operated in the condition received for 1 hour with the orifice wide open. Each class B and C cleaners shall be operated in the condition received for a period of 8 hours with the orifice wide open. After this run all accessible lubrication points shall be visually checked. Any evidence of excessive leaking, galling, or lack of lubrication shall be cause for rejection. The commutator(s) shall be visually examined. There shall be no evidence of excessive burning, pitting, or grooving.

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4.3.2.2.3 Noise tests. The cleaner shall be set up in operational configuration with hose, wand, and bare floor or combination tool supplied with the cleaner attached on type I, or with the agitator operating on type II and III cleaners. The tests shall be taken with the cleaner cleaning an ASTM plush test carpet placed upon an ASTM test pad that conforms with ASTM Standard F655. With the cleaner in operation, two (2) readings shall be taken using the A weighted, damped, or slow response scale. A third reading shall measure the background noise after shutdown. On type I cleaners the readings shall be taken 6 feet from the hose connection, 6 feet from the floor, with the wand being held in the operating position pointing away from the cleaner and the attached tool resting on the floor. On type III cleaner the test shall be the same as with the type I cleaner with the agitator in operation. On the type II cleaner the readings shall be taken 4 feet behind the cleaner, 6 feet above the floor with the handle in a position approximately where an operator would be holding it. If the difference between the average of the first two readings and the background reading is greater than 14 db, disregard the background reading. If less than 14 db difference, combine the readings logarithmically. The average or log sum (as applicable) shall not be greater than 85 dbA.

4.3.2.2.4 Performance tests. Each sample cleaner shall be tested in accordance with ASTM Standard F558. No taping sealing of the hose or wand connections shall be permitted. The cleaners shall demonstrate the capability to develop the minimum water lift as specified in 3.4.3 and 3.5.3 as applicable. If the type II vacuum cleaner being tested is not required to have the specified attachments, and does not have the capability of attaching a hose for the performance tests, the tests shall be performed with a plate that seals off the nozzle of the cleaner while not interfering with the rotation of the agitator, and has an opening to allow the attachment of a hose to be run to the plenum chamber.

4.4 Inspection for preparation and delivery. An inspection shall be made to determine whether the packaging, packing, and marking comply with the requirements of the contract or purchase order. Examination for defects shall be in accordance with table I. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

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Table I
Preparation for delivery defects.

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Any component missing, damaged, or not conforming to applicable requirements stated herein.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, or distortion of container.
Contents	Quantity of container is more or less than required. Gross weight exceeds requirements of the box specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The vacuum cleaners shall be packaged and packed in accordance with industry standards and shall be in compliance with the contract or purchase order. Marking shall be in accordance with the contract or purchase order.

6. NOTES

6.1 Intended use. These vacuum cleaners are intended for general housekeeping and carpet maintenance and are to be used for dry pickup only.

6.2 Ordering data. The purchasers should select the preferred options permitted herein and include the following in procurement documents.

(a) Title, number, and date of this specification.
(b) Type, class, and size desired (if applicable) (see 1.2).
(c) If tools are required for type II (see 3.5.6).
(d) Voltage required (see 1.1).
(e) Packaging, packing, and marking required (see section 5).

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6.3 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing employees to undue hazards from the recovered materials.

6.4 Reference NSN's. The following NSN's use this document as the primary reference:

7910-00-550-9123	Type I, Class A, 115 VAC
7910-00-479-6684	Type I, Class B, 115 VAC
7910-00-550-9124	Type II, Class A, 115 VAC w/ Accessories
7910-00-550-9125	Type II, Class A, 115 VAC
7910-00-720-5536	Type II, Class B, 115 VAC
7910-00-468-4197	Type II, Class C, Size 1, 115 VAC
7910-01-154-1151	Type II, Class C, Size 2, 115 VAC
7910-01-158-6628	Type III, Class A, 220 VAC

MILITARY INTERESTS

NONE: DoD has waived coordination on revisions and amendments to this Federal Specification until further notice.

Preparing Activity
GSA-FSS/7FXEG

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